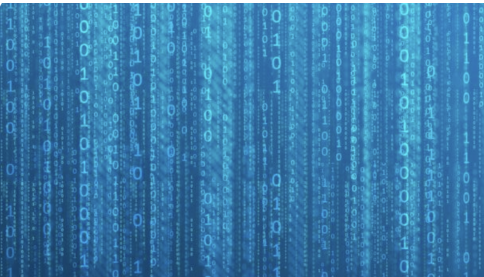


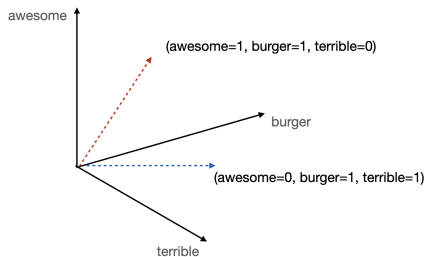
Data Representation

Procheta Sen



Representing Categorical Data

If you must represent categorical data, then you could do so by representing each category as a separate dimension of a vector.



Data Representation

- ▶ Data representation is one of the first things we must do in data mining.
- ▶ What we can mine is largely determined by our data representation.
- ▶ There is no one best data representation method for all data mining tasks.

Representing Text Data: Example

In what ways can we represent the following sentence?

Example Sentence: 'The burger I ate was an awesome burger!'

- ▶ Method 1: By a list of words: ['the', 'burger', 'i', 'ate', 'was', 'an', 'awesome', 'burger']
- ▶ Method 2: By the set of words: {'the', 'burger', 'i', 'ate', 'was', 'an', 'awesome'}
- ▶ Method 3: By a vector of word frequency ('the':1, 'burger':2, 'i':1, 'ate':1, 'was':1, 'an':1, 'awesome':1)
- ▶ Method 4: By a vector of letter frequency: {'a': 4, ' ': 7, 'b': 2, 'e': 6, 'g': 2, 'i': 1, 'h': 1, 'm': 1, 'o': 1, 'n': 1, 's': 2, 'r': 4, 'u': 2, 't': 2, 'w': 2}